

Closed Conversation: Woven City – Is It a City?



Image of Toyota Woven City (BIG, 2025)

Mysterious yet lively, contrasting with other cities around. When you see this picture, you do not realize the surrounding environment exists, as if it were sleeping. But Woven City is alive and in operation.

In the 2020 Consumer Electronics Show (CES), Woven City was introduced as an urban incubator and living laboratory project for future mobility and sustainability. At the 2025 CES, Toyota reintroduced its phase progress and vision, providing a five-year update. Woven City recently completed the first phase (2021 to 2024), and started welcoming its first residents, approximately 350 people, mainly Woven City workers and their families. In the future, their goal is to have 2,000 people residing in the city, including innovators, partners, and workers.

Toyota Woven City is a site of 175 acres (about 708,000 square meters or 70.8 hectares) – one-fifth of Central Park– developed at the former Higashi-Fuji plant of Toyota Motor East Japan site in Susono City, Shizuoka Prefecture. Toyota Motor’s Higashi-Fuji plant was closed in 2020 due to the relocation of production lines, and its labor was transferred to Iwate and Miyagi. The manufacturing plant site has now been moved to Toyota’s prototype smart city and mobility testbed (Toyota Times, 2025).



Higashi-Fuji plant of Toyota Motor East Japan site in Susono, Shizuoka Prefecture, Japan (Toyota Times, 2025)

Layers of Woven City Initiatives

Woven City, an experimental, fully corporate-led smart city beneath Mount Fuji, has captured global and academic attention. It exemplifies **smart urbanism, data solutionism, and platform urbanism** through its vision of a **living laboratory built on real-world data**. Woven City's vision, operational mechanisms, design, and governance can be critically examined through the frameworks of smart urbanism, the urbanity of AI, and data solutionism.

In 2018, the former CEO of Toyota Motors Group, Akio Toyoda, claimed Toyota was transforming from an automobile maker into a **mobility company**. The website of Woven City proclaims its mission and purpose as *"Woven City as a mobility test course and co-creation," "developing and demonstrating mobility of people, goods, information, and energy"* (Woven by Toyota, 2025). The mission emphasizes the development of mobility technology while demonstrating a big shift in its business model from **hardware-first to software-first**.

The research done by Akabane, an economist, adds context to this shift into a mobility company. He highlighted the urgency of Toyota's **weak position in the growing Battery Electric Vehicle (BEV)** market, which facilitated the transformation into a mobility company and the launch of Woven City (Akabane, 2024). Woven City could be Toyota's **business strategy** instead of rushing into competition in the BEV market.

The Woven City's mechanism illustrates this shift and its connection to Toyota's evolving business model. Designed to enable inventors to co-create products and services using Toyota's software and technologies, Woven City aims to drive the **development of new ideas for the next generation of mobility**. This includes the movement of people, goods, information, and energy, all supported by the infrastructure and services provided by both Woven City and Toyota. Toyota Group's official articles describe it as a human-centered city to *"unlock new possibilities"* and pursue *"well-being for all"* by *"Inventors and Weavers"* (Global Toyota, 2025). The infrastructure and services in Woven City are primarily designed to facilitate and contain **technological development**, creating an especially invention-friendly environment. One of the main infrastructure centers is Woven Inventor Garage, which was an automobile manufacturer's plant from 1967 until 2020, and is now part of Woven City. Toyota's introduction of the transformation of the old structure into Woven City emphasized connecting the past to the future, keeping the memory of the Higashi-plant (Toyota Times, 2025). Woven City reflects Toyota's adaptation to a changing business landscape, but this transformation is **strategically wrapped in symbolic narratives that serve both branding and innovation agendas**.

"Designed as a human-centered city, Woven City places the people who gather there—Inventors and Weavers—at its center to unlock new possibilities in the shared pursuit of "Well-being for All"(Global Toyota, 2025).



Woven Inventor Garage under **renovation** (Toyota Times, 2025)

Although Woven City refers to itself as a “**city**,” the language surrounding its introduction suggests a different identity. The use of the word “**launch**,” typically reserved for products, brands, or services, frames Woven City more as a **commercial innovation than a civic project**. Its presentations at CES in further reinforces this view. CES is a global platform known for showcasing technological advancements and consumer-ready solutions, not urban planning models or community-based initiatives. In this context, Woven City appears less as a city where people settle and form communities, and more as a controlled testbed for Toyota to prototype, evaluate, and refine its next-generation mobility technologies in line with its evolving business strategy.

*“Toyota Motor Corporation (“Toyota”) and Woven by Toyota, Inc. (“WbyT”) today announced the official **launch of Toyota Woven City (“Woven City”)**, first unveiled at CES 2020 and jointly developed by the two companies.(Toyota, 2025)”*

Beneath its aspirational vision, Woven City stands as a testament to the **urbanity of AI** (Cugurullo et al., 2024) and a **material manifestation of smart urbanism** shaped by embedded data solutionism. At its core, the city is designed to test and implement artificial intelligence (AI) across both the built and virtual environments—from **road systems and autonomous vehicles to transportation management and smart homes**. A key design element is a new version of a typical tree-type road system, presented as a new model of mobility equality between pedestrians and vehicles. This system divides the **ground level** into three exclusive lanes: one each for pedestrians, autonomous vehicles, and pedestrians using personal mobility devices (e.g., scooters, bicycles), all orchestrated by AI technologies for continuous testing (Pegoraro, 2020). Below ground, additional infrastructure supports autonomous delivery and logistics, insulated from environmental conditions (Woven City, 2025). These dual-layered systems reveal the city’s function not as a space for daily life, but as a **platform optimized for experimenting** with next-generation mobility under controlled, AI-mediated conditions.

“With people, buildings, and vehicles all connected and communicating with each other through data and sensors, we will be able to test AI technology in both the virtual and the physical world, maximizing its potential,” Toyota said (Pegoraro, 2020).

Urbanity of AI, data-solutionism, and Woven City

The scale of Woven City, an estimated \$10 billion project, raises critical questions about the extent to which **AI development now demands physical urban infrastructure**. The project reinforces Cugurullo et al.’s (2024) argument that “AI needs the city” to evolve its intelligence. In a 2021 interview with Toyota Times, former CEO Akio Toyoda emphasized that **experiments with AI are only possible “in**

the form of a city,” pointing to the necessity of immersive, real-world conditions for meaningful data acquisition. This framing recasts the city not as a space for public life, but as a training ground for algorithmic learning, where **urban experience is deliberately curated to optimize machine intelligence**. In this model, the **city becomes a tool** purposefully-built to generate feedback loops for proprietary technologies, rather than serve diverse public needs.

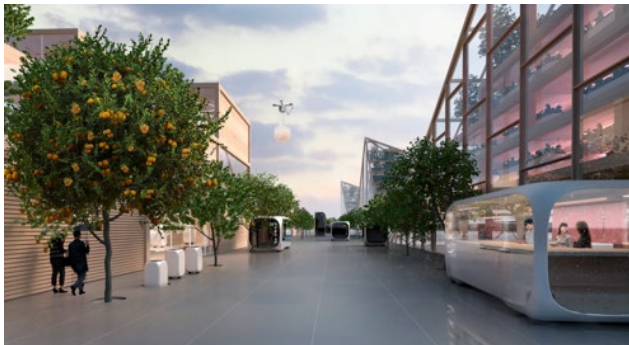
*“Woven City is a test course for Toyota to become a mobility company. **This experiment can only be done in the form of a city**” (Pegoro, 2020)*



Pedestrian and slow vehicles road (BIG, 2025)



Pedestrian only road (BIG, 2025)



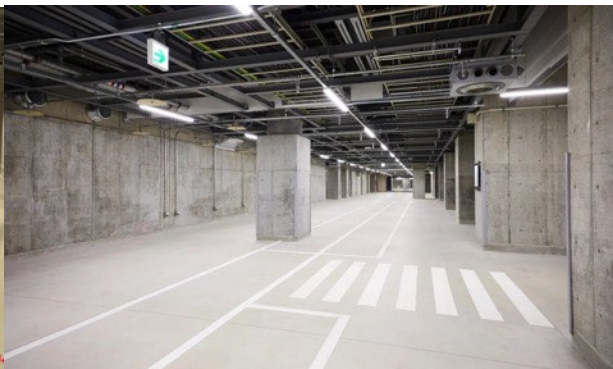
Pedestrians' road with parked autonomous vehicles (BIG, 2025)



Exclusive autonomous vehicles road (Youtube, 2025)



Underground level of Woven City (YouTube, 2025)

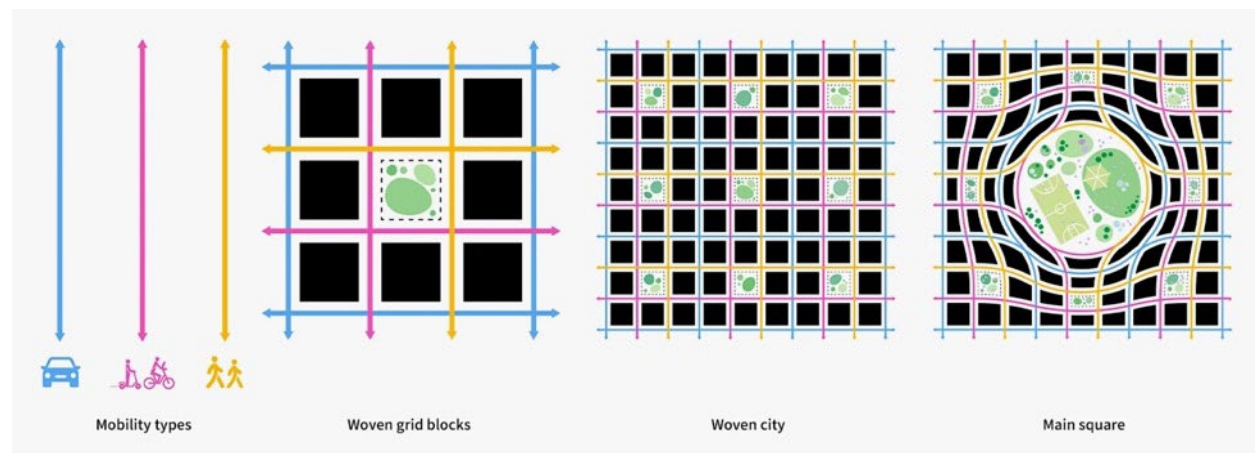


Underground parking space (Woven City, 2025)

Woven City positions **data** as the key to realizing the future of mobility and artificial intelligence. Its vision implies that next-generation mobility technologies can solve complex urban problems and deliver “well-being for all.” Implicitly, **this narrative reframes the promise: it is not just mobility, but data itself can bring a social progress.** This logic aligns with what Mattern (2013) terms **data solutionism** as a belief that entangled, multidimensional social realities can be **reduced to computable problems** with optimal, technical solutions. The emphasis on testing and data collection within Woven City is central to its design and purpose. As Mattern warns, such an approach turns the **city into a machine for generating, processing, and optimizing data.** In Woven City, this machinic logic not only informs its infrastructure but extends to its design, governance and external relations, **redefining the relationships** among people, built environment, and technology.

Design, A new spatial logic

The design of Woven City was developed in collaboration with the international architecture firm Bjarke Ingels Group (BIG), and is structured around the “Woven Grid”, a 3×3 layout of blocks, each measuring 150 meters in width. Peripheral roads are allocated for vehicles, while inner pathways are reserved for pedestrians and cyclists. Each block is defined by eight surrounding buildings and a central courtyard, with differentiated architectural and social functions envisioned to foster diverse micro-neighborhoods. Woven Grid has three different modes of mobility pass through and pockets of social space (ecograda, 2018).



Concept of Woven Grid (ecograda, 2024)



Conceptual plan of Woven City (ecogradia, 2024)

BIG emphasized “**equity**” among three different modes of mobility: automobile, pedestrian, and bike. However, **this language of equity invites critique: it implies a parity of value between human bodies and autonomous machines, raising questions about the priorities embedded in such spatial logics.** The infrastructure does not merely accommodate mobility, but orchestrates it under algorithmic governance, where movement becomes part of a measurable system.

Residential buildings are designed in a **terraced, stepped form** that ascends toward Mount Fuji, blending symbolic orientation with spatial hierarchy. The design reinforces the significance of its location by orienting the city toward Mount Fuji, a culturally sacred and symbolically powerful landmark in Japan. This spatial alignment embeds a narrative of reverence and continuity, layering the city’s technological ambitions with a gesture toward national identity and spiritual heritage. This orientation and narrative are a gesture to facilitate public acceptance and justification of Woven City through design.



Woven Grid (Woven by Toyota, 2025)



Residential Building and Mountain Fuji (BIG, 2025)

Woven City and the Rise of Fully Owned Private Urbanism

Woven City represents a unique smart city model in which a single private actor—Toyota—designs, operates, and owns the entire system, infrastructure, and data. Featuring corporate leadership and funding, technological integration and innovation hubs, it is categorized to be **corporate smart city** (Kato & Ito, 2024). Built on privately owned land, Woven City exemplifies a fully corporate-led urban project, expanding the typology of corporate smart cities such as **Songdo** in South Korea and **Masdar** in Abu Dhabi (Kato & Ito, 2024). Unlike public-private partnerships, this model centralizes power in the hands of a private entity, reflecting the growing ambition of tech corporations to govern urban systems through proprietary platforms and data-driven solutions (Ito & Kato, 2024). Viewed through Sadowski's (2021) framework, which traces smart urbanism through three escalating phases: **oversight of governance, operation of services, and ownership of space**. Woven City emerges as a definitive case of the third phase: **full corporate sovereignty**.

Toyota exercises comprehensive control over Woven City's property, services, and data infrastructure, effectively transforming urban space into a closed, proprietary system. Sadowski cautions against this final phase of smart urbanism, where **private tech firms not only mediate urban life but own its core structures**. By contrast, **Songdo City aligns more with the first phase**, where companies like Cisco provide the digital backbone to monitor and support urban governance, but do not directly own or operate the city itself. While Songdo integrates smart infrastructure and aims to attract international business and residents, it remains formally **embedded within public governance structures** of taxation, democratic administration, and open access to housing and services. Woven City, therefore, marks a critical shift from **public interest governance to corporate territoriality in the name of innovation**.

Players and Writers of Woven City – Governance, Relationship, Experiences

If a corporation oversees, operates, and owns the governance of a city, then who and what is being governed? Understanding the definition of the public realm and the degree of openness is essential to evaluating whether Woven City functions as a city or not. The “public” within Woven City is pre-defined and restricted to two categories: **“Inventors” and “Weavers.”** These roles are not open to all, but curated by Toyota, shaping a closed ecosystem optimized for data generation and feedback.

Inventors are corporate affiliates, startup teams, or project groups developing products and services aligned with Toyota's vision of future mobility. ***Weavers***, on the other hand, are residents or visitors who live in the city and engage with these technologies, effectively acting as test users. The two groups form a deliberate, algorithmic feedback loop: **weavers generate behavioral data that informs and improves the inventors' technologies**. Crucially, both groups are **selectively admitted** by Toyota, reinforcing a model of **controlled participation** where public life is redefined as data production. Viewed through the lens of smart urbanism, this setup renders the social structure of Woven City not organic, but engineered with residents instrumentalized as inputs. In this model, citizenship takes a functional role in a privately governed urban experiment.

Surprisingly, weavers are selected applicants and visitors to Woven City. The access is controlled and restricted to the public, even the citizens of Susuno City, revealing the closeness of Woven City.

Disconnection with Susuno City in terms of governance and interaction of population, is often discussed in research interviews questioning its legitimacy of Smart City. Despite Woven City's ambitious technological vision, **interviews with local stakeholders reveal a growing sense of disconnection between Toyota's project and the local context of Susuno City**. According to Ito and Kato (2024), local residents and Susuno City officials expressed limited involvement in the planning and governance of Woven City, despite its physical location within their municipal boundaries. In fact, city officials acknowledged that they were only marginally informed about Toyota's progress, and **no formal administrative integration** or shared governance structure was established between the city and the Woven project. This governance gap raises critical questions about the legitimacy of Woven City as a “city”, especially considering its strict access control, non-residential status of most users, and limited application of legal frameworks for public participation. Toyota representatives interviewed rationalized the closed nature of the city by emphasizing the need for data privacy and controlled experimentation environments. However, this justification only underscores the corporate logic that governs the project that Woven City exists for data and private interest, but not for public participation. As a result, Woven City becomes an enclave of private sovereignty, rather than a public urban entity embedded in local civic

life. This disconnection not only exposes the limits of private urbanism but also puts risk of integration and with the surrounding population and environment.

In conclusion, Toyota's Woven City leaves question as a city, **representing a radical redefinition of what a city is and who it is for. Branded as a futuristic urban prototype**, Woven City is not governed by public institutions or civic engagement, but by corporate ownership and control. It is a closed, territorialized space built by Toyota to collect, manage, and experiment with data in a tightly controlled environment. The elaboration on vision and promotion of Woven City diffuses unspoken initiatives of business strategy and AI technology development under controlled environment. Unlike open cities that permits democratic governance and public participation, Woven City **selectively invites** "Weavers"—employees, developers, and collaborators—to live, work, and generate data within its boundaries, ruling private ownership. The city functions more as a machine or a live-in laboratory, where every activity is captured, monitored, and turned into input for future Toyota products and services.

In Woven City, “city” is rather a concept of **creating a boundary for physical enclosure** which allows for the **territorialization of data**—where data is not openly shared, but privatized. This closed-loop system transforms the city into a container of information that becomes a territory of its own, echoing the logic of control and optimization rather than public life. While today's cities are becoming ever more networked, Woven City reflects a deeper shift: from infrastructure for citizens to infrastructure for data, from public-led city to private-owned city. Yet, as John Sexton (Mattern, 2013) reminds us, cities are more than just infrastructure and technology; they are also built on social interactions, culture, and neighborhoods. Without public participation and freewill of citizens, Woven City is closer to a branded theme park of technological experimentation, that borrows a “city” for justification.

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